Dataset Documentation

Telecom NetOps AI Assistant — 100,000-record dataset

1. Dataset Name

5G Network Operations (NetOps) Dataset. The project uses the local CSV file **5G_netops_data_100K.csv**, reported by the project run output as containing 100,000 records and 27 columns.

2. Dataset Source

The project specification identifies the primary source as the 5G NetOps Dataset hosted on Hugging Face by the “fenar” account. The project's download script reports downloading the dataset from Hugging Face and creating `data/raw/5G_netops_data_100K.csv`. The exact upstream revision and file checksum are not established by the available project output.

3. Dataset Link

Primary: https://huggingface.co/datasets/fenar/5g_netops
Alternative reference listed in the project specification: https://github.com/wineslab/RANalyzer-Dataset

4. Data Format

CSV (Comma-Separated Values), stored at `data/raw/5G_netops_data_100K.csv`. Reported size: 100,000 rows × 27 columns. The dataset is tabular and includes numeric network measurements plus categorical and geographic context.

5. Key Fields

The following field names are based on the project specification and sample row; exact capitalization and punctuation may vary in the source CSV.

Field	Meaning / use
Season	Seasonal context
Cell Availability (%)	Availability
MTTR (hours)	Mean time to repair/restore
Throughput (Mbps)	Network throughput
Latency (ms)	Network latency
Packet Loss Rate (%)	Packet loss
Call Drop Rate (%)	Call-drop rate
Handover Success Rate (%)	Handover success
Alarm Count	Total alarms
Critical Alarm Count	Critical alarms
Parameter Changes	Network parameter changes
Successful Configuration Changes (%)	Configuration-change success
Data Usage (GB)	Data usage
User Count	User count
Signal Strength (dBm)	Radio signal strength
Jitter (ms)	Jitter
Connection Setup Success Rate (%)	Connection setup success
Security Incidents	Security incident count

Field	Meaning / use
Authentication Failures	Authentication failure count
Temperature (°C)	Environmental temperature
Humidity (%)	Humidity
Weather	Weather condition
Issue Reported	Issue indicator/category
City	City metadata
State	State metadata
Zip	Postal code
Fault Occurrence Rate (%)	Fault occurrence metric

The UI also displays incident identifiers such as NETOPS-000410. Confirm in the source code whether this identifier is a CSV field or is attached/generated by the application; it is not among the 27 fields listed above.

6. Dataset Purpose

The dataset supplies historical telecom network performance and operational context to the Telecom NetOps AI Assistant. It is used to retrieve and rank similar historical cases for natural-language queries; support metadata filtering, incident classification and prioritization; inform fault-risk and performance analysis; ground root-cause hypotheses and troubleshooting recommendations; populate the Dataset Explorer; and support retrieval evaluation.

The records are tabular observations. Without a usable timestamp or ordered time-series field, they do not by themselves establish whether a metric is increasing or decreasing over time.

7. Data Preprocessing Performed

The following steps are supported by the project context and observed runtime output:

• Dataset acquisition: the download script retrieves the dataset from Hugging Face and writes data/raw/5G_netops_data_100K.csv.

• CSV loading: the application successfully reported loading 100,000 rows and 27 columns.

• Schema use: the application reads and displays network KPI, alarm, security, environmental and location fields.

• Retrieval preparation: the application builds or loads a retrieval index/cache for the configured 100,000-row dataset; runtime logs reported retrieval_index_cache=hit rows=100000.

• Query-time retrieval and ranking: matching records are ranked for a query, while the UI's top-results control limits displayed results without necessarily limiting the underlying dataset scope.

Verification note: Exact missing-value handling, duplicate removal, type conversion, outlier treatment, normalization/scaling, categorical encoding, embedding model/version, index persistence format, and whether the CSV itself is modified were not confirmed from the available evidence. These should only be described as performed after checking the corresponding code and file. This document does not propose changing the 100,000-record dataset.